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1 Math

1.1 Vectors

$$a(\bar{A} + \bar{B}) = a\bar{A} + a\bar{B}$$

$$\bar{A} \cdot \bar{B} = |\bar{A}||\bar{B}|\cos(\theta) \quad (1.1)$$

$$\bar{A} \times \bar{B} = |\bar{A}||\bar{B}|\sin(\theta)\hat{n} \quad (1.4)$$

$$(\bar{A} \times \bar{B}) = -(\bar{B} \times \bar{A}) \quad (1.6)$$

$$\bar{A} \cdot (\bar{B} \times \bar{C}) = \bar{B} \cdot (\bar{C} \times \bar{A}) = \bar{C} \cdot (\bar{A} \times \bar{B}) \quad (1.15)$$

$$\bar{A} \times (\bar{B} \times \bar{C}) = \bar{B}(\bar{A} \cdot \bar{C}) - \bar{C}(\bar{A} \cdot \bar{B}) \quad (1.17)$$

$$(\bar{A} \times \bar{B}) \cdot (\bar{C} \times \bar{D}) = (\bar{A} \cdot \bar{C})(\bar{B} \cdot \bar{D}) - (\bar{A} \cdot \bar{D})(\bar{B} \cdot \bar{C})$$

$$\bar{A} \times [\bar{B} \times (\bar{C} \times \bar{D})] = \bar{B}[\bar{A} \cdot (\bar{C} \times \bar{D})] - (\bar{A} \cdot \bar{B})(\bar{C} \cdot \bar{D}) \quad (1.18)$$

1.2 Calculus

$$dT = \left(\frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z} \right) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z})$$

$$= (\nabla T) \cdot d\vec{l}, \quad (1.35)$$

where the gradient (∇) is an operator:

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}. \quad (1.39)$$

The gradient is used to calculate the divergence of a vector with:

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}, \quad (1.40)$$

and the curl with:

$$\nabla \times \bar{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z}. \quad (1.41)$$

The product rules for the gradient are:

$$\nabla(fg) = f\nabla g + g\nabla f$$

$$\nabla(\bar{A} \cdot \bar{B}) = \bar{A} \times (\nabla \times \bar{B}) + \bar{B} \times (\nabla \times \bar{A}) + (\bar{A} \cdot \nabla)\bar{B} + (\bar{B} \cdot \nabla)\bar{A}$$

$$\nabla \cdot (f\bar{A}) = f(\nabla \cdot \bar{A}) + \bar{A} \cdot (\nabla f)$$

$$\nabla \cdot (\bar{A} \times \bar{B}) = \bar{B} \cdot (\nabla \times \bar{A}) - \bar{A} \cdot (\nabla \times \bar{B})$$

$$\nabla \times (f\bar{A}) = f(\nabla \times \bar{A}) + \nabla f \times \bar{A}$$

$$\nabla \times (\bar{A} \times \bar{B}) = \bar{A}(\nabla \cdot \bar{B}) - \bar{B}(\nabla \cdot \bar{A}) + (\bar{B} \cdot \nabla)\bar{A} - (\bar{A} \cdot \nabla)\bar{B}$$

When we use the gradient to make a second derivative we get the Laplacian:

$$\nabla(\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \quad (1.42)$$

and the laplacian of a vector:

$$\nabla^2 \bar{v} = (\nabla^2 v_x) \hat{x} + (\nabla^2 v_y) \hat{y} + (\nabla^2 v_z) \hat{z}. \quad (1.43)$$

We also have the following identities:

$$\nabla \times (\nabla T) = 0 \quad (1.45)$$

$$\nabla \cdot (\nabla \times \bar{v}) = 0 \quad (1.46)$$

$$\nabla \times (\nabla \times \bar{v}) = \nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}. \quad (1.47)$$

1.3 Fundamental theorem of calculus

Gradient:

$$\int_a^{\bar{b}} (\nabla T) \cdot d\bar{l} = T(\bar{b}) - T(\bar{a}) \quad (1.55)$$

Divergence:

$$\int_V (\nabla \cdot \bar{v}) dV = \oint_S \bar{v} \cdot d\bar{a} \quad (1.56)$$

Curl:

$$\int_S (\nabla \times \bar{v}) \cdot d\bar{a} = \oint_P \bar{v} \cdot d\bar{l} \quad (1.57)$$

2 Electrostatics

2.1 Electric field

The force on a test charge (Q) from a single point charge (q), a distance $\bar{r}$ away, is calculated by Coulomb's law:

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ is the permittivity of free space.